



COMSATS University Islamabad, Attock Campus

Lab Task

Data Structure

Student Name	Abdullah
Registration Number	FA25-BSC-015
Date	Sep 11,2026
Program	BSCS
Semester	III

Instructor: Mr. Bilal Bukhari

Lab Task 01

Code:

```
int main()
{
    int x, start, stop;
    int sum=0;    cout<<"Enter X:";
    cin>>x;

    cout<<"Enter starting value:";    cin>>start;

    cout<<"Enter stopping value:";    cin>>stop;

    for(int i=start;i<=stop;i++)
    {
        sum= sum+x*x;
    }

    cout<<"Sum="<<sum;    return 0;
}
```

Result

```
Enter X:4
Enter starting value:2
Enter stopping value:5
Sum=64
-----
Process exited after 11.68 seconds with return value 0
Press any key to continue . . . |
```

Lab Task 02

Code:

```
#include <iostream> using namespace std;

class ArrayList { private:
    int arr[100]; int size;

public:
    ArrayList() { size = 0;
    }

    // 1. Insert at end void insertEnd(int
val) { if (size < 100) {
arr[size] = val; size++; cout <<
"Inserted " << val << " at end\n";
    } else { cout << "List full!\n";
    }
    }

    // 2. Insert at start void insertStart(int val) { if (size <
100) { for (int i = size; i > 0; i--) arr[i] = arr[i - 1];
arr[0] = val; size++; cout << "Inserted " << val << "
at start\n";
```

```

    } else {        cout << "List full!\n";
    }
}

```

```

// 3. Insert after specific value    void
insertAfter(int after, int val) {    int pos = -1;
for (int i = 0; i < size; i++) {    if (arr[i] ==
after) {        pos = i;        break;
    }
    if (pos == -1) {        cout << after << " not
found!\n";        return;
    }    if (size < 100) {        for (int i = size; i > pos + 1; i--)
arr[i] = arr[i - 1];        arr[pos + 1] = val;        size++;        cout <<
"Inserted " << val << " after " << after << endl;
    } else {        cout << "List full!\n";
    }
}

```

```

// 4. Insert before specific value    void
insertBefore(int before, int val) {    int pos = -1;
for (int i = 0; i < size; i++) {    if (arr[i] == before) {
pos = i;        break;
    }
    if (pos == -1) {        cout << before << " not
found!\n"; return;
    }    if (size < 100) { for (int i =
size; i > pos; i--)
arr[i] = arr[i - 1];
arr[pos] = val;        size++;
cout << "Inserted " << val <<
" before " << before << endl;
    } else {        cout << "List full!\n";
    }
}

```

```

// 5. Display void display() { if (size ==
0) { cout << "List is empty!\n";
return;
}

```

```

    cout << "List: "; for (int i = 0; i <
size; i++) cout << arr[i] << " ";
cout << endl;
}

```

```

// 6. Delete from end void deleteEnd() {
if (size == 0) { cout << "List empty!\n";
return;
}

cout << "Deleted " << arr[size - 1] << " from end\n";
size--;

}

```

```

// 7. Delete from start void deleteStart() {
if (size == 0) { cout << "List empty!\n";
return;
}

cout << "Deleted " << arr[0] << " from start\n"; for (int i =
0; i < size - 1; i++) arr[i] = arr[i + 1]; size--;
}

```

```

// 8. Delete specific value void
deleteVal(int val) { int pos = -1; for
(int i = 0; i < size; i++) { if (arr[i] ==
val) { pos = i; break;
}
}

if (pos == -1) { cout << val << " not
found!\n"; return;
} for (int i = pos; i < size - 1; i++) arr[i] = arr[i
+ 1]; size--;
cout << "Deleted " << val << endl;

}

```

```

// Lab Task 3: Linear Search using while loop    void
linearSearch(int val) {
    int i = 0;    int found = 0;

    while (i < size) {        if (arr[i] == val) {        cout << val <<
" found at index " << i << endl;        found = 1;        break;
    }        i++;
    }

    if (found == 0)        cout << val << " not
found!\n";
    }
};

```

```

int main() {    ArrayList list;    int
choice, val, target;

    do {
        cout << "\n===== MENU =====" << endl;

        cout << "1. Insert at end" << endl;        cout << "2. Insert at start"
<< endl;        cout << "3. Insert after specific value" << endl;        cout
<< "4. Insert before specific value" << endl;        cout << "5. Display
list" << endl;        cout << "6. Delete from end" << endl;        cout <<
"7. Delete from start" << endl;        cout << "8. Delete specific value"
<< endl;        cout << "9. Linear Search (while loop)" << endl;
        cout << "10. Exit" << endl;        cout << "Choice: ";        cin >> choice;
    }
}

```

```

switch (choice) {        case 1:
    cout << "Enter value: ";        cin >>
val;        list.insertEnd(val);        break;
        case 2:
    cout << "Enter value: ";        cin >>
val;        list.insertStart(val);        break;
        case 3:
    cout << "Enter after value: ";
    cin >> target;        cout << "Enter new
value: ";        cin >> val;
    list.insertAfter(target, val);        break;
}

```

```

        case 4:
            cout << "Enter before value: ";          cin >>
target;      cout << "Enter new value: ";
cin >> val;    list.insertBefore(target, val);
break;

        case 5:
list.display();    break;
        case 6:
list.deleteEnd();    break;
        case 7:
list.deleteStart();    break;
        case 8:      cout << "Enter value to
delete: ";          cin >> val;
list.deleteVal(val);    break;
        case 9:
            cout << "Enter value to search: ";          cin >>
val;      list.linearSearch(val);          break;

        case 10:      cout << "Exiting...\n";
break;

        default:
            cout << "Invalid choice!\n";

    }
} while (choice != 10);

return 0;
}

```

Result

```
C:\Users\Laraib\Documents\C  X  +  v

===== MENU =====
1. Insert at end
2. Insert at start
3. Insert after specific value
4. Insert before specific value
5. Display list
6. Delete from end
7. Delete from start
8. Delete specific value
9. Linear Search (while loop)
10. Exit
Choice: 1
Enter value: 2
Inserted 2 at end

===== MENU =====
1. Insert at end
2. Insert at start
3. Insert after specific value
4. Insert before specific value
5. Display list
6. Delete from end
7. Delete from start
8. Delete specific value
9. Linear Search (while loop)
10. Exit
Choice: |
```

Lab Task 03

Code:

```
#include <iostream>    using
namespace std;

int main() {    int arr[100], size, search,
i;
```

```

    cout << "Enter number of elements: ";    cin >>
size;

    cout << "Enter " << size << " elements: ";    for (i =
0; i < size; i++)    cin >> arr[i];

    cout << "Enter value to search: ";    cin >>
search;

    i = 0;    while (i < size && arr[i] !=
search)

        i++;    if (i < size)    cout << search << " found at
index " << i << endl;    else    cout << search << " not
found!" << endl;

    return 0;
}
Result

```

```

Enter number of elements: 4
Enter 4 elements: 2 3 5 7
Enter value to search: 5
5 found at index 2

-----
Process exited after 20.25 seconds with return value 0
Press any key to continue . . . |

```

